Lab 1 — Example: Notebooks & Markdown

ITCS 3162 — Introduction to Data Mining

This walkthrough introduces the environment you'll use for the rest of the course: Jupyter notebooks (and Google Colab), Python cells, and Markdown for documenting your work.

Learning objectives

By the end of this notebook you will be able to:

1. Identify the difference between a code cell and a markdown cell
2. Run cells and understand execution order
3. Write formatted markdown (headings, lists, code, math, links, images)
4. Use basic keyboard shortcuts to work efficiently

> **Tip:** read each markdown cell, then run the code cell below it. Pay attention to the `[ ]` brackets to the left of code cells — the number inside tells you the execution order.

1. Code cells

A code cell runs Python. Click the cell below and press **Shift + Enter** to run it.

```
# This is a code cell. Anything after # is a comment.
print("Hello, data mining!")
2 + 2
```

```
Hello, data mining!
4
```

Notice two things:

- The `print()` output appears below the cell.
- The last expression (`2 + 2`) is also displayed — notebooks automatically show the value of the last line.

Execution order matters

Variables persist between cells in the order you run them, not the order they appear in the notebook. Run the next two cells and watch the `[N]` counter.

```
x = 10
y = 5
```

```
x + y
```

```
15
```

If you ever feel "lost" in a notebook, use **Kernel ⟶ Restart & Run All** (Jupyter) or **Runtime ⟶ Restart and run all** (Colab) to start fresh from the top.

2. Markdown cells

Markdown cells render formatted text. Double-click this cell to see the raw markdown, then press **Shift + Enter** to render it again.

Headings

Use `#` for headings. More `#`s = smaller heading.

```
# H1 — Lab title
## H2 — Section
### H3 — Subsection
```

Emphasis

- `*italic*` ⟶ *italic*
- `**bold**` ⟶ **bold**
- `` `inline code` `` ⟶ `inline code`

Lists

Unordered:

- apples
- bananas
    - cavendish
    - plantain
- cherries

Ordered:

1. Load data
2. Explore
3. Model

Links and images

- A link: UNC Charlotte
- An image: `![alt text](https://example.com/image.png)`

Code blocks

Triple backticks, optionally with a language for syntax highlighting:

```
import pandas as pd
df = pd.read_csv("data.csv")
```

```
df.head()
```

Math (LaTeX)

Inline: `$y = mx + b$` renders as $y = mx + b$.

Display:

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y_i})^2$$

Tables

Column	Type	Example
age	int	24
name	str	Alice
active	bool	True

3. Useful keyboard shortcuts

Press `Esc` to enter **command mode** (blue cell border), then:

Key	Action
`A`	Insert cell **above**
`B`	Insert cell **below**
`D D`	**Delete** cell (press D twice)
`M`	Convert to **markdown** cell
`Y`	Convert to **code** cell
`Z`	Undo cell deletion

Press `Enter` to go back to **edit mode** (green border).

To run a cell: **Shift + Enter** (run and move to next) or **Ctrl/Cmd + Enter** (run and stay).

4. Colab vs. local Jupyter

Both run notebooks; small differences:

Feature	Jupyter (local)	Google Colab
Setup	Install Anaconda/Python	Just a Google account
Files	Local filesystem	Google Drive, uploads
GPU	Your machine	Free GPU/TPU available
Packages	`pip install` in terminal	`!pip install ...` in a cell

In Colab, prefix shell commands with `!`:

```
# This works in both Colab and Jupyter
!python --version
```

```
Python 3.12.3
```

5. A first taste of data

We'll dive into pandas properly in Lab 3, but here's a one-cell preview so you can see why notebooks are a good fit for data work.

```
import pandas as pd

# A tiny inline dataset
df = pd.DataFrame({
    "student": ["Alice", "Bob", "Carla", "Dan"],
    "score":   [88, 72, 95, 64],
    "passed":  [True, True, True, False],
})
df
```

	student	score	passed
0	Alice	88	True
1	Bob	72	True
2	Carla	95	True
3	Dan	64	False

Notice how the DataFrame renders as a nicely formatted table — that's the notebook detecting a pandas object and rendering it as HTML.

Summary

You now know how to:

- Run code cells and follow execution order
- Write markdown with headings, lists, code, math, and tables
- Use shortcuts to move quickly
- Tell Colab and Jupyter apart

Move on to `lab_01_exercise.ipynb` to practice on your own.